

COGS AND CARRACKS

The French used galleys for cross-Channel raiding and in the battles of the Hundred Years War—the oared vessels were often supplied by Genoese mercenaries. The Castilians also used galleys successfully against the English, who for their part had balingers, small singled-masted vessels powered by a combination of oars and sails. But ships powered by sails alone were becoming increasingly important in northern wars. Sturdy single-masted cogs were the trusty workhorses of maritime trade in the Baltic and North Sea. With high castles added fore and aft, these merchant ships provided excellent platforms for the archers who played a major role in combat at sea, as well as spacious holds for the purposes of military transport.

The carrack emerged in the 15th century as a large hybrid of the cog and sleeker Mediterranean lateen-sailed ships. The Genoese brought carracks into the Channel in support of the French fighting England's Henry V in 1416. Henry himself built the massive *Grace Dieu*, a carrack as large as a first-rate ship of the line of the Nelson era. Such "Great Ships" appealed to monarchs for their prestige value. Initially, their fighting advantage lay primarily in how difficult they were to board, with their high inward-sloping hull. But by the start of the 16th century they were being equipped with cannon in large numbers, with gun ports cut into the hull. Purpose-built warships such as England's *Mary Rose* were a new kind of fighting machine whose firepower would make earlier forms of naval warfare redundant.

Portuguese carracks
Having explored the entire west coast of Africa, the Portuguese reached their true goal when Vasco da Gama completed his historic voyage to India in 1498. Carracks were the mainstays of their long-distance voyages of exploration.